

WARRENTON-FAUQUIER, VA, USA

WMO: 722692

Lat: 38.586N

Lon: 77.711W

Elev: 103

StdP: 100.09

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00367

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.0	-19.1	0.7	-4.3	-16.6	0.9	-2.9	10.4	0.9	9.1	1.9	1.0	300	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	35.2	24.5	33.7	23.9	32.4	23.4	27.5	32.4	26.5	30.4	25.4	29.2	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.9	30.0	25.2	20.6	28.7	24.4	19.6	27.8	88.0	33.4	83.1	30.3	78.2	28.8	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	5.7	-15.1	37.1	3.5	1.9	-17.6	38.5	-19.7	39.6	-21.7	40.6	-24.3	42.0		
(5)			-15.4	27.4	3.5	2.1	-18.0	28.9	-20.0	30.1	-22.0	31.2	-24.6	32.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.9	1.2	4.2	7.5	13.9	19.0	23.3	25.6	24.3	20.9	14.3	7.6	4.1	(6)
(7)		DBStd	9.49	5.38	5.44	5.61	4.95	4.35	2.99	2.84	2.69	3.74	4.68	4.59	4.99	(7)
(8)		HDD10.0	894	277	176	117	17	1	0	0	0	12	98	196		(8)
(9)		HDD18.3	2391	531	396	337	152	47	2	0	17	144	322	443		(9)
(10)		CDD10.0	2310	5	14	40	133	280	399	483	443	328	146	27	12	(10)
(11)		CDD18.3	764	0	1	2	18	68	151	225	186	95	19	1	0	(11)
(12)		CDH23.3	7811	0	5	35	248	685	1503	2411	1831	897	190	6	1	(12)
(13)		CDH26.7	3306	0	0	7	76	234	611	1167	797	360	54	0	0	(13)
(14)	Wind	WSAvg	1.9	2.2	2.4	2.7	2.5	1.9	1.6	1.3	1.3	1.5	1.8	1.9	1.9	(14)
(15)	Precipitation	PrecAvg	1048	75	55	97	83	103	101	103	83	104	80	81	81	(15)
(16)		PrecMax	1433	168	166	229	126	208	213	180	185	259	205	157	146	(16)
(17)		PrecMin	746	25	15	0	40	35	35	34	31	0	2	22	40	(17)
(18)		PrecStd	157	40	31	50	27	47	45	41	36	66	51	41	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	23.9	27.3	32.0	32.9	36.1	37.9	36.0	35.0	30.8	23.9	21.3	(19)
(20)		0.4%	MCWB	14.9	15.3	19.1	19.0	23.3	26.4	27.6	24.5	22.2	21.9	16.7	18.6	(20)
(21)		2%	DB	16.5	19.9	23.1	28.9	31.2	33.4	36.1	33.9	32.9	27.9	21.1	18.2	(21)
(22)		2%	MCWB	12.7	13.5	15.2	18.2	22.9	24.4	25.4	23.4	22.6	20.4	15.3	16.1	(22)
(23)		5%	DB	13.0	16.8	20.9	26.3	29.5	32.1	34.1	32.6	31.0	25.1	18.5	15.7	(23)
(24)		5%	MCWB	9.0	11.3	14.6	17.5	21.9	23.2	24.8	23.2	21.7	18.1	13.8	12.8	(24)
(25)		10%	DB	10.0	13.0	17.8	23.5	27.5	30.4	32.5	31.3	28.4	22.8	16.3	12.6	(25)
(26)		10%	MCWB	7.0	8.8	12.6	15.9	20.8	22.5	24.0	23.0	20.4	17.9	12.1	9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	16.6	20.9	23.1	26.8	27.9	29.3	27.5	27.1	23.6	19.1	19.4	(27)
(28)		0.4%	MCDWB	18.8	20.7	24.0	28.3	31.7	33.9	35.9	32.3	31.7	28.2	21.1	20.5	(28)
(29)		2%	WB	13.0	14.7	17.7	21.0	24.6	26.5	28.2	26.3	25.2	22.1	17.1	17.3	(29)
(30)		2%	MCDWB	15.9	19.2	20.3	25.9	28.6	31.1	33.5	29.5	28.5	26.0	19.5	18.5	(30)
(31)		5%	WB	9.6	11.7	15.4	18.9	23.2	25.3	27.0	25.3	23.7	20.7	15.1	13.4	(31)
(32)		5%	MCDWB	12.5	15.8	20.2	23.7	27.4	29.5	31.3	29.2	27.7	23.7	17.6	15.1	(32)
(33)		10%	WB	7.2	9.3	12.9	17.0	21.9	23.9	25.7	24.3	22.6	18.9	12.5	10.3	(33)
(34)		10%	MCDWB	9.8	13.0	17.1	22.1	26.0	28.2	29.3	28.8	25.9	22.2	15.3	11.9	(34)
(35)	Mean Daily Temperature Range	MDBR	10.9	11.5	12.8	14.0	12.4	12.3	12.3	12.5	12.6	13.5	12.8	10.9	(35)	
(36)		5% DB	MCDBR	15.5	16.4	17.1	17.3	14.7	14.0	14.3	14.1	16.0	16.1	16.7	13.8	(36)
(37)		5% DB	MCWBR	11.5	10.9	10.1	8.3	7.4	6.3	5.5	5.3	6.8	8.7	10.9	11.0	(37)
(38)		5% WB	MCDBR	13.0	14.2	14.4	14.2	12.7	12.4	12.2	11.3	11.8	12.7	12.7	12.3	(38)
(39)		5% WB	MCWBR	10.6	10.3	10.3	8.1	7.0	6.6	5.8	5.4	6.2	8.0	9.8	10.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.301	0.317	0.352	0.393	0.432	0.490	0.505	0.507	0.418	0.356	0.319	0.305	(40)	
(41)		taud	2.498	2.432	2.392	2.302	2.218	2.096	2.076	2.052	2.281	2.454	2.513	2.520	(41)	
(42)		Ebn at Noon	883	910	907	885	853	801	784	771	826	854	852	850	(42)	
(43)		Edh at Noon	78	95	109	127	141	160	162	162	121	92	77	72	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.13	2.95	3.92	4.90	5.51	6.07	5.90	5.29	4.35	3.31	2.41	1.87	(44)	
(45)		RadStd	0.19	0.33	0.26	0.37	0.56	0.44	0.32	0.26	0.52	0.37	0.20	0.18	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+72

Nomenclature: See separate page